

Header lines

DN 10 / DN 15

Installation and operating instructions



English (GB) Installation and operating instructions

Original installation and operating instructions.

CONTENTS

	Page		
1. Safety instructions	2	7.2 Retrofitting old header lines	12
1.1 Symbols used in this document	2	7.2.1 Assembly with Loctite	12
1.2 Safety advice for handling chlorine	2	7.2.2 Assembly with hemp	12
1.2.1 Hazards of chlorine gas	2	7.3 Maintenance of connection valve	13
1.2.2 Hazards of liquid chlorine	3	7.4 Maintenance of the shut-off valve	14
1.3 Qualification of the users	3	8. Spare parts	14
1.4 Responsibilities of the owner	3	9. Fault finding	14
1.5 Maintenance and service personnel	3	9.1 Leakage at the connection valve	14
1.6 Protective equipment	3	9.2 Leakage at the shut-off valve	14
1.6.1 Personal protective equipment	3	10. Decommissioning	15
1.6.2 Obligations of the operating authority	3	10.1 Short-term decommissioning (up to 2 weeks)	15
1.6.3 Storage of the protective equipment	3	10.2 Long-term decommissioning (more than 2 weeks)	15
1.7 First aid in case of accidents	3	11. Disposal	15
1.7.1 First aid after inhaling chlorine	3		
1.7.2 First aid after chemical burn of skin	3		
1.7.3 First aid after chemical burn of eyes	3		
1.7.4 First aid after internal chemical burn	3		
1.8 Handling chlorine containers	4		
1.8.1 Steel cylinders	4		
1.8.2 Chlorine drums	4		
1.8.3 Basic rules	4		
1.8.4 Valid regulations	4		
1.9 Chlorine rooms	4		
1.9.1 Regulations for chlorine rooms	4		
1.9.2 Labelling of chlorine rooms	5		
2. Product description	5		
2.1 Correct usage	5		
2.2 Inappropriate usage	5		
2.3 Identification	5		
2.3.1 Nameplate	5		
2.4 Warranty	5		
3. Technical data	6		
3.1 Header line for chlorine gas cylinders	6		
3.1.1 Connections	6		
3.1.2 Weights	6		
3.1.3 Materials	6		
3.2 Header line for chlorine gas drums	6		
3.2.1 Connections	6		
3.2.2 Weights	6		
3.3 Header line for ammonia and sulphur dioxide drums	6		
3.3.1 Connections	6		
3.3.2 Weights	6		
3.4 Dimensions	7		
3.4.1 Header line for 3 cylinders	7		
3.4.2 Header line for 5 drums	7		
3.5 Physical and chemical data of chlorine	8		
4. Construction and function	8		
4.1 Description of the device	8		
4.2 Function	8		
5. Startup	9		
5.1 Installation	9		
5.1.1 Assembling the header line	9		
5.1.2 Tightening the connection valve	9		
5.2 Checking the tightness	10		
5.2.1 Checking the tightness of a large system	10		
5.2.2 Checking the tightness of parts leading chlorine gas or sulphur dioxide	10		
5.2.3 Checking the tightness of parts leading ammonia	11		
5.3 Start-up	11		
6. Operation	11		
6.1 Container exchange	11		
7. Maintenance	12		
7.1 Replacing the flexible lines	12		
7.1.1 Flexible copper lines for chlorine	12		
7.1.2 Flexible stainless-steel lines for ammonia and sulphur dioxide	12		

**Warning**

Prior to installation, read these installation and operating instructions. Installation and operation must comply with local regulations and accepted codes of good practice.

1. Safety instructions

These installation and operating instructions contain important information for the user of the product:

- Technical data
- Instructions for commissioning and use
- Safety instructions

If you require further information, or if problems occur that are not described in detail in this manual, please contact Grundfos.

1.1 Symbols used in this document

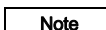
This manual contains the following standardised safety instructions about possible residual risks:

**Warning**

If these safety instructions are not observed, it may result in personal injury.

**Caution**

If these safety instructions are not observed, it may result in malfunction or damage to the equipment.

**Note**

Notes or instructions that make the job easier and ensure safe operation.

1.2 Safety advice for handling chlorine

Country-specific local laws, regulations, and standards apply for the handling of chlorine.

1.2.1 Hazards of chlorine gas**Warning**

Toxic by inhalation.

Irritating to eyes, respiratory system and skin!

- Causes whooping cough, dyspnoea and lacrimation.
- Has a slight paralysing effect to the central nervous system.
- Concentrations of more than 10 ppm chlorine gas in the respiratory air mean an acute danger to life.
- Inhaling air with a high concentration of chlorine gas for a long time is lethal.

1.2.2 Hazards of liquid chlorine

- Causes burns of skin.
- Causes reddening and blistering of skin.
- Rules of conduct
- Change chlorine containers only with gas mask.
- Enter contaminated rooms only with protective suit and compressed-air respirator.
- In case of flight wear gas mask, if possible. Observe wind direction.

1.3 Qualification of the users

Based on his professional training and experience, the user must have a good knowledge of the operation of chlorine gas dosing systems. He must be so familiar with the relevant laws and regulations governing safety in the workplace and accident prevention, and with the recognised standards of good practice, that he is able to judge the safe working condition of a chlorine gas dosing system.

The user is responsible for operation and monitoring of the product at the installation site.

The responsibilities of the user:

- Observe the safety data sheets from the chemicals supplier.
- Be trained by qualified Grundfos personnel in the operation of the product
- Observe the relevant regulations governing safety in the workplace and accident prevention
- Wear appropriate protective clothing in accordance with national regulations for the prevention of accidents when operating the system and handling chemicals.

1.4 Responsibilities of the owner

The operator of the product is responsible for the following:

- Ensure that this manual is kept clearly accessible in the immediate vicinity of the system for the entire service life.
- Meet the installation requirements specified by the manufacturer (environmental conditions, required water connections and fittings, electrical connection, protective pipe for dosing line if necessary, audible or optical warning device if necessary).
- Ensure that water lines and armatures are regularly checked, serviced and maintained.
- Obtain official approval for storing chemicals, if necessary.
- Train users in the operation of the system.
- Ensure that the regulations for the prevention of accidents are observed in the installation site.
- Provide all users and service personnel with protective clothing (e.g. face mask, gloves, protective apron) in accordance with the national regulations.

1.5 Maintenance and service personnel

The product may only be maintained and serviced by personnel authorised by Grundfos.

1.6 Protective equipment

1.6.1 Personal protective equipment

The operating authority of a chlorine gas dosing system has to provide respiratory equipment (full-sight gas mask), personally fitted, labelled by name, with an effective chlorine filter (B2P3) and at least 1 spare filter per gas mask, for each operator.

The protective equipment must be stored easily accessible outside the chlorine rooms.

- For systems with chlorine drums at least 2 protective suits with compressed-air respirators must be available.

1.6.2 Obligations of the operating authority

- Introducing the operators to handling the protective equipment.
- The operators must use the personal protective equipment regularly, or carry out exercises with it at least every 6 months.
- The gas mask filters must be replaced regularly
 - after the expiry of the date of durability
 - at least 6 months after opening (note the opening date on the filter)
 - after contact with chlorine.
- Observe employing prohibition according to the local laws!

1.6.3 Storage of the protective equipment

- Outside the chlorine rooms
- Well visible
- Easily available at any time
- Protected from dust and moisture.

1.7 First aid in case of accidents

1.7.1 First aid after inhaling chlorine

1. Keep calm.
2. Remove injured persons from the dangerous area.
3. Helpers must pay attention to personal protection.
4. Remove contaminated clothes.
5. Calm down injured persons, and keep them warm with blankets.
 - Supply fresh air, use oxygen respirator, if possible.
 - No mouth-to-mouth resuscitation!
6. Call medical aid or transport to hospital
 - lying
 - sitting in case of difficulty in breathing.
 - State chlorine as the cause.

1.7.2 First aid after chemical burn of skin

1. Keep calm.
2. Remove contaminated clothes.
3. Rinse skin with plenty of water.
4. Bandage the wound aseptically.
5. Seek medical aid.
 - State chlorine as the cause.

1.7.3 First aid after chemical burn of eyes

1. Keep calm.
 - Rinse eyes with plenty of water while the patient is lying.
 - Protect healthy eye, if necessary.
 - Spread eyelids widely, let the eye move to all sides.
2. Seek ophthalmologist.
 - State chlorine as the cause.

1.7.4 First aid after internal chemical burn

1. Keep calm.
2. Drink water in short sips.
 - If possible, take medical charcoal.
3. Seek medical aid.
 - State chlorine as the cause.

1.8 Handling chlorine containers

Chlorine is stored in grey steel cylinders or drums in lockable chlorine rooms. Due to safety precautions, chlorine containers are only filled up to 95 % of their capacity.

Kind of gas, weight, owner, producing date and date of the last testing have to be noted clearly on the container.

Observe the following:

- Store chlorine containers fireproof.
- Protect chlorine containers from heat and sunlight.
- Chlorine containers should not be modified or repaired by the user.
- Keep full and empty containers closed.

1.8.1 Steel cylinders

- Contents: 50 kg or 65 kg
- Extract chlorine gas from the upright standing cylinder via the valve.

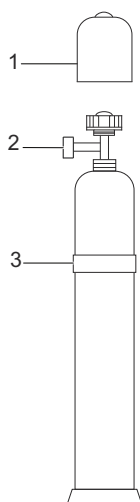


Fig. 1 Design of a cylinder

Pos.	Description
1	Protective cap
2	Valve
3	Holding clip for securing the cylinder

1.8.2 Chlorine drums

- Contents: 500 kg or 1000 kg
- Extraction of chlorine gas via the riser pipe valve.
- Extraction of liquid chlorine via the dip pipe valve.

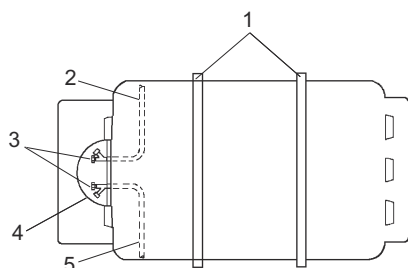


Fig. 2 Design of a drum

Pos.	Description
1	Riser pipe for chlorine gas extraction
2	Valve for chlorine gas extraction
3	Valve for liquid chlorine extraction
4	Dip pipe for liquid chlorine extraction
5	Tyre
6	Mark for adjustment of drum

1.8.3 Basic rules



Warning

Handling of chlorine containers only by experienced personnel.

Observe safety regulations for chlorine containers.

- Treat containers carefully, do not throw!
- Protect containers from turning over or rolling away, e.g. with chains or clips.
- Protect containers from direct sunlight and temperatures over 50 °C.
- Transport containers only with valve protection nut and protective cap.



Warning

These rules apply for both full and empty containers, as empty containers still contain rests of chlorine and therefore are under pressure.

1.8.4 Valid regulations



Warning

Local laws and regulations for handling, transport and storage of chlorine must be observed!

- Regulations for accident prevention with process instructions.
- Regulations concerning places of work
- Technical rules for pressurized gases
- Chlorine gas dosing systems for water treatment
- Chlorine for the treatment of water for the human use (EN 937)

1.9 Chlorine rooms

Chlorine rooms are rooms, where a chlorine gas dosing system and/or chlorine containers are located. The chlorine in these rooms is under pressure.

1.9.1 Regulations for chlorine rooms

According to the German regulations for accident prevention "Chlorination of water", chlorine rooms have to meet the following requirements:

- They should not have connections to other rooms, and must be separated gas-tight and fire-resistant.
- They must be at ground level, dry and air-conditioned. The recommended room temperature is 18-20 °C. It should be neither lower than 0 °C nor higher than 50 °C.
- Overpressure lines of dosing equipment must not end in the open air.
- Ventilation openings leading into the open air must be limited to 2 x 20 cm².
- Appropriate exhaust devices with downstream adsorption systems must be installed.
- Chlorine gas must not get into lower-lying rooms, shafts, pits, canals or aspirating holes of ventilation systems.
- Chlorine rooms must be equipped with a water sprinkling system to precipitate the escaping chlorine gas.
 - The sprinkling system must have a run-off with air trap.
 - Operation must be possible manually from outside the chlorine rooms.
- A chlorine gas warning system must be installed
 - with optical and acoustical alarm device,
 - with connection to the water sprinkling system,
 - with a warning system that reactivates automatically after switch-off (e.g. for container exchange).

TM04 8943 1813

TM04 8940 1813

- The floor of a chlorine room must be even, the exit not below ground level and not higher than a possible loading ramp.

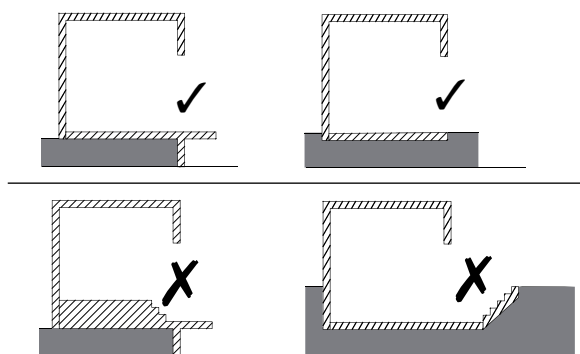


Fig. 3 Requirements of chlorine rooms

- Chlorine rooms must have a direct exit to the open air.
- Doors must be lockable, open outwards, and it must be possible to open the doors without a key from inside the room.
- Chlorine rooms must not be dedicated for the permanent stay of people.
- Eating, drinking, or storing food in chlorine rooms is prohibited

1.9.2 Labelling of chlorine rooms

In Germany, the following warning signs must be installed outside the entrance of a chlorine room:



Fig. 4 Warning sign and supplementary sign "Chlorination plant: Access only for instructed persons"

Inside a chlorine room, the mandatory sign "Use the gas mask" and the "Instruction sheet for first aid in case of chlorine gas intoxications" must be installed.



Fig. 5 Mandatory sign

2. Product description

2.1 Correct usage

Header lines connect several chlorine gas cylinders or chlorine gas drums with the components of a gas dosing system.

Header lines must be used exclusively for the applications described in this manual and with the gas they are approved for. The usage is limited to industry, water treatment and waste water treatment.

2.2 Inappropriate usage

Operational safety is only guaranteed, if the product is used correctly. All operating methods conflicting with correct usage are not permitted, and lead to the expiry of all liability claims.



Warning

Unauthorised structural modifications to the product may result in serious damage to equipment and personal injury.

2.3 Identification

The header line can be identified by the connection type of the T-piece (cemented or welded), and the material of the flexible line (copper or stainless steel).

Identification of the header line

T-piece	cemented	header line for cylinders
	welded	header line for drums
Flexible line	copper	header line for chlorine gas
	stainless steel	header line for ammonia or sulphur dioxide gas

2.3.1 Nameplate

The indications on the nameplate are:

- product number
- serial number

2.4 Warranty

A guarantee claim in accordance with our general terms of sale and delivery is only valid if the following requirements are fulfilled:

- The product is used in accordance with the information within this manual.
- The installation, startup, operation and maintenance is carried out by authorised and qualified staff.

3. Technical data

Caution

The values stated in the technical data must be adhered to.

3.1 Header line for chlorine gas cylinders

The vacuum regulator is directly connected to the DN 15 header line. Depending on the chosen header line, up to 8 cylinders can be connected.

Line pressure	max. 16 bar
Ambient temperature	0 °C to +40 °C
Capacity of gaseous chlorine	max. 200 kg/h
Capacity of liquid chlorine	max. 600 kg/h

3.1.1 Connections

Cylinder connection: 1", G 3/4, G 5/8, 1.03"-14, M 26x3, 1 1/4", closed yoke (USA), 15/16-8 RH, G 1/2

Vacuum regulator connection: 1", G 3/4, M 26x3

3.1.2 Weights

Number of cylinders	1*	2	3	4	5	6	7	8
Weight [kg]	1	3.4	5.1	6.8	8.5	10.2	11.9	13.6

* without collector pipe (only connection valve and copper line)

3.1.3 Materials

Collector pipe	Seamless drawn steel pipe, painted, T-piece screwed in and cemented
Flexible lines	Copper, tin-plated
Valve body	Brass, nickel-plated
Valve piston	Hastelloy C
Gasket of pressure connection and gasket of copper pipe	Aramid fibres with NBR
Packing	PTFE
O-rings	FKM

3.2 Header line for chlorine gas drums

DN 15 and DN 25 header lines collect gas or liquid from several drums. Depending on the chosen header line, up to 8 drums can be connected.

Line pressure	max. 16 bar
Ambient temperature	0 °C to +40 °C

3.2.1 Connections

Drum connection: 1", G 3/4, G 1/2, G 5/8, M26x3, 1 1/4", 1.03"-14, closed yoke (USA), 15/16-8 RH

Connection flange of outlet: DN 15 or DN 25

3.2.2 Weights

Number of drums	Weight [kg]	
	DN 15	DN 25
1*	1.3	1.3
2	11	17
3	13.8	21
4	18	27.5
5	21	31
6	23.5	34.5
7	28	41
8	31	44.5

* without collector pipe (only connection valve and copper line)

Materials

Collector pipe	Seamless drawn steel pipe, painted, connections welded
Flexible lines	Copper, tin-plated
Body of connection valve	Brass, nickel-plated
Body of shut-off valve	Grey cast iron, painted
Valve piston	Hastelloy C
Connection gaskets	Aramid fibres with NBR
Packing	PTFE
O-rings	FKM

3.3 Header line for ammonia and sulphur dioxide drums

DN 15 and DN 25 header lines collect gas or liquid from several drums. Depending on the chosen header line, up to 8 drums can be connected.

Line pressure	max. 16 bar
Ambient temperature	0 °C to +40 °C

3.3.1 Connections

Pressure connection: 1", G 3/4, G 5/8, W 21.8x1/14

Connection flange of outlet: DN 15 and DN 25

3.3.2 Weights

Number of drums	Weight [kg]	
	DN 15	DN 25
1*	1.4	1.4
2	11.5	17.5
3	14.3	21.5
4	18.5	28
5	22	32
6	24.5	35.5
7	29	42
8	32	45.5

* without collector pipe (only connection valve and copper line)

Materials

Collector pipe	Seamless drawn steel pipe, painted, connections welded
Flexible lines	Stainless steel 1.4571
Body of connection valve	
Body of shut-off valve	Grey cast iron
Valve piston	Hastelloy C
Connection gaskets	Aramid fibres with NBR
Packing	PTFE
O-rings	EPDM

3.4 Dimensions

3.4.1 Header line for 3 cylinders

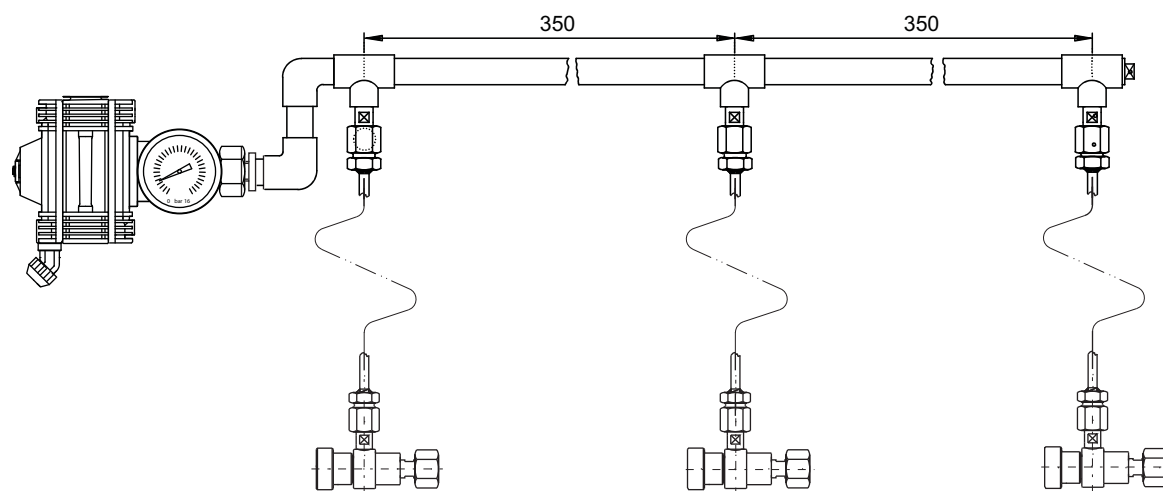


Fig. 6 Dimensions of header line for 3 cylinders

3.4.2 Header line for 5 drums

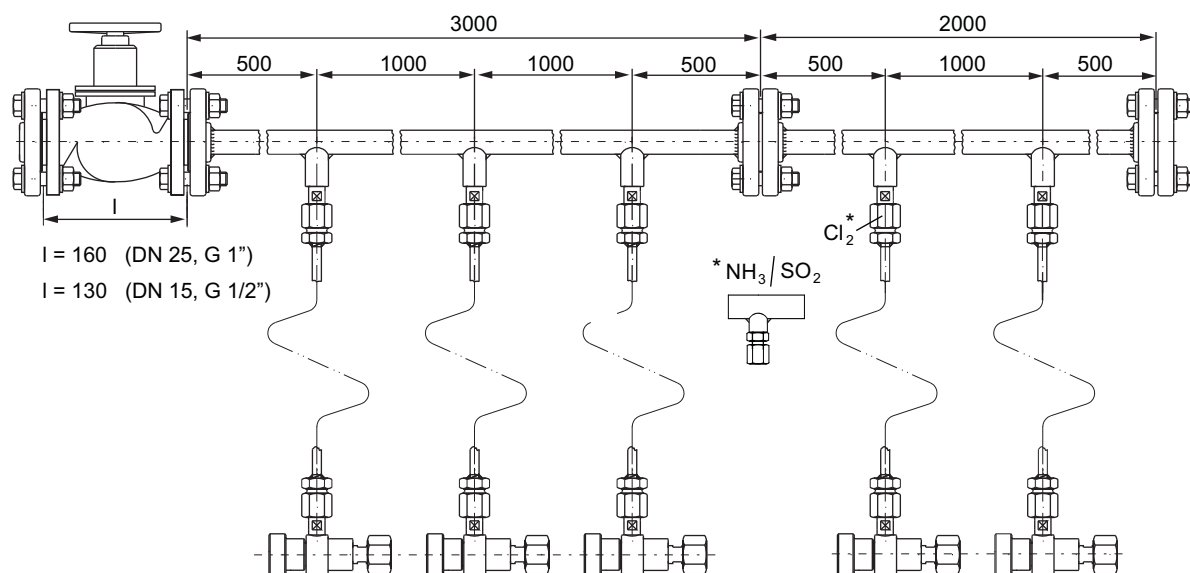


Fig. 7 Dimensions of header line for 5 drums

TM04 8637 4112

TM04 8608 4112

3.5 Physical and chemical data of chlorine

Under normal conditions of pressure and temperature, chlorine is a yellowish green gas with a pungent odour, which exists as Cl₂ molecule.

It is not flammable, but can promote the flammability of metals, hydrocarbons, etc.

Atomic weight	35.453 u
Molecular weight	70.906 u
Boiling point	-34 °C
Melting point	-101 °C
Degree of purity according to EN 937	99.5 %
TLV (Threshold Limit Value)	0.5 ml/m ³ (1.5 mg/m ³)

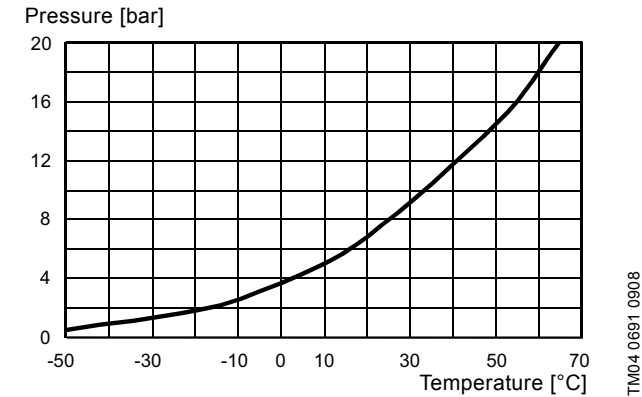


Fig. 8 Vapour pressure curve of chlorine

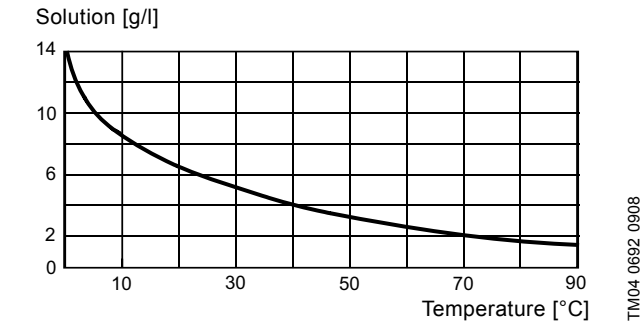


Fig. 9 Solubility of chlorine gas in water

4. Construction and function

The installation location must be protected from direct sunlight and rainfall.

Caution *The admissible temperatures must be observed and excessive temperature variations must be avoided.*

4.1 Description of the device

A header line connects several gas containers with the components of the gas dosing system. A header line consists of connection valves (5) for connection of the container to the flexible line (3), a T-piece (2) and a rigid collector pipe (1). The connection valves (5) can be closed in case of container exchange.

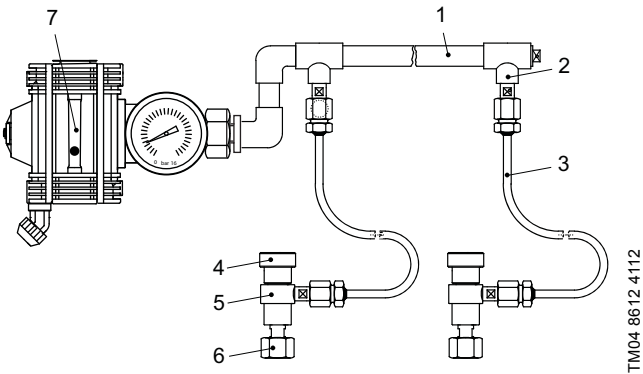


Fig. 10 Assembled header line

Pos.	Description
1	Collector pipe
2	T-piece
3	Flexible line
4	Lock nut
5	Connection valve
6	Connection for the gas container
7	Vacuum regulator

4.2 Function

When several gas containers are connected to a header line, the gas extraction quantity in the gas dosing system increases by the number of connected containers.

The maximum extraction quantity must not exceed 1 % of the container filling quantity. Otherwise the gas will liquefy during extraction.

If a gas container is empty, it can be exchanged. Before container exchange, the connection valve must be closed by fastening the lock nut (4) as far as it will go.

5. Startup

Header lines are a part of a gas dosing system. Before start-up, the tightness of header lines must be checked together with the corresponding gas dosing system.

5.1 Installation

5.1.1 Assembling the header line

After unpacking, assemble the header line strainless as soon as possible, in order to avoid penetration of humidity. Observe that no foreign matter gets into the header line.

1. Fix the collector pipe (1) to the wall with the enclosed pipe clips.
2. Connect the vacuum regulator (7) to the collector pipe (1).
3. Place the cylinders or drums correctly and fix them. Fix cylinders with support clips.
4. Screw the flexible lines (3) to the T pieces (2) of the collector pipe.
5. Insert the gaskets into the connection valves (4). Screw the connection valves to the cylinder valves.
6. Screw the flexible lines (3) to the connection valves (4).

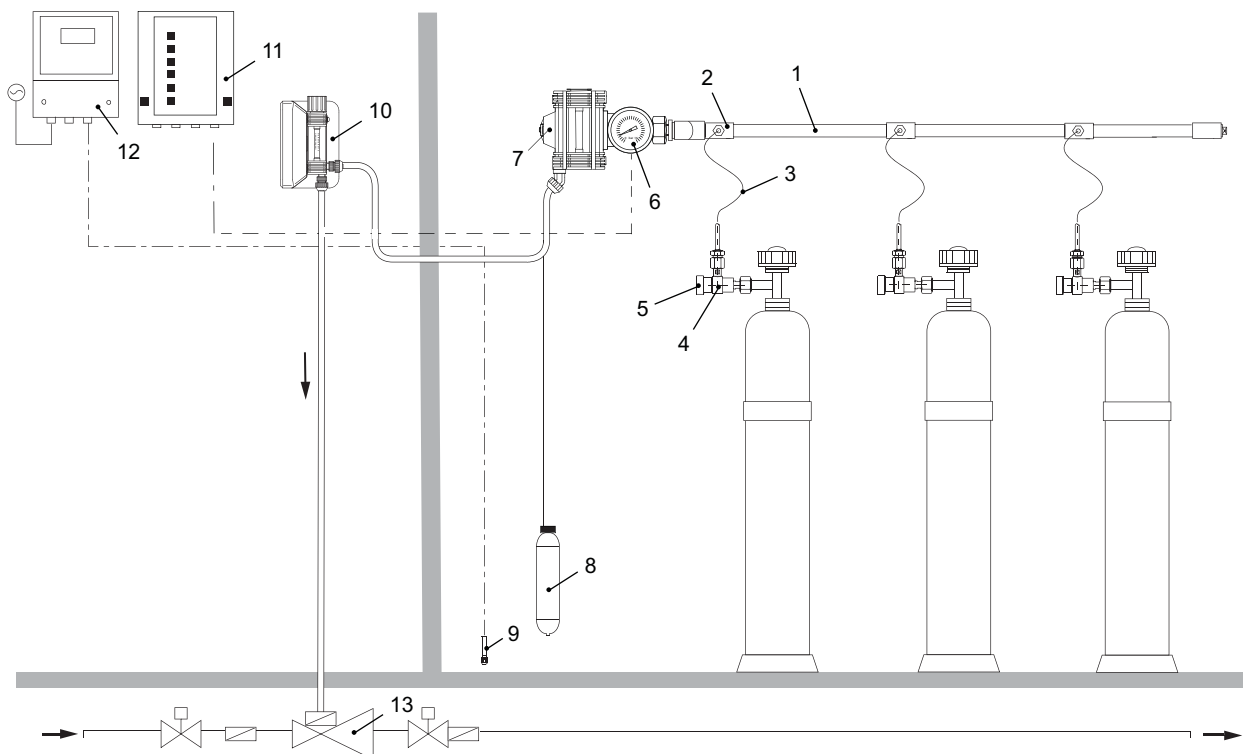


Fig. 11 Gas dosing system with header line

Pos.	Description
1	Collector pipe
2	T-piece
3	Flexible line
4	Connection valve
5	Lock nut
6	Pressure gauge
7	Vacuum regulator
8	Adsorption cylinder
9	Gas sensor
10	Dosing regulator
11	Empty indication display unit
12	Gas warning device
13	Water apparatus with injector

5.1.2 Tightening the connection valve

Before start-up of the header line, the connection valves must be tightened.

1. Unscrew the lock nut (1) of the connection valve
2. Tighten the gland (3) with a torque of 15 Nm.
3. The installation dimension from the spindle end to the valve body must be 43.5 mm.

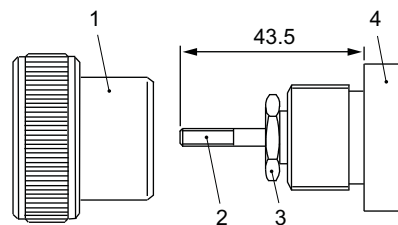


Fig. 12 Connection valve: installation dimension of the spindle

Pos.	Description
1	Lock nut
2	Spindle
3	Gland
4	Valve body

4. Screw in the lock nut (1) as far as it will go. The connection valve is closed.

TM04 8611 4112

TM04 8610 4112

5.2 Checking the tightness

After an exchange of containers or flexible copper lines, the tightness of the header line must be checked.

1. The tightness of large systems can be checked with a nitrogen rinsing device.
2. The tightness of parts leading chlorine gas or sulphur dioxide can be checked with ammonia. The tightness of parts leading ammonia can be checked with sulphur dioxide.

5.2.1 Checking the tightness of a large system

This is a rough tightness check, which can only be done with a nitrogen rinsing device.

Note

In addition, the fine checking has to be done independently.

Checking the tightness with nitrogen

1. Close all chlorine container valves.
2. Open container connection valves and all shut-off valves up to the chlorine gas dosing system.
3. Open the connection valve of the nitrogen cylinder.
4. Slowly open the valve of the nitrogen cylinder, until the lines have a pressure of about 10 bar (read at the pressure gauge of the vacuum regulator).



Warning

Maximum nitrogen pressure 16 bar.

Danger of damages and leakages when being exceeded.

5. Apply soap water to all components under pressure.
 - If bubbles form and/or pressure drops at the pressure gauge, the pressure lines are leaky.
6. Depressurise the system.
7. Eliminate the leakage.
8. Repeat the tightness check.
9. If no bubbles form, and the pressure at the pressure gauge does not drop significantly within one hour, the pressure lines are tight.

5.2.2 Checking the tightness of parts leading chlorine gas or sulphur dioxide

Even very small leakages become evident with ammonia vapour, which reacts with chlorine gas forming white mist.



Warning

Ammonia is very toxic for water organisms! Liquid ammonia must not come in contact with lines or parts of the system, because it is very corrosive.

Remove drops immediately with a dry cloth.

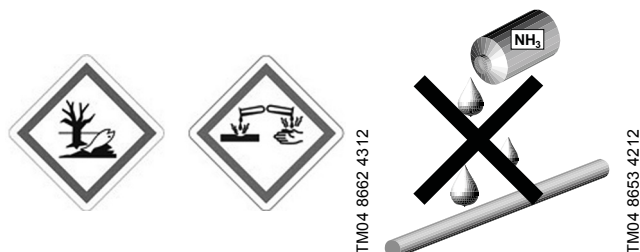


Fig. 13 Handling of ammonia



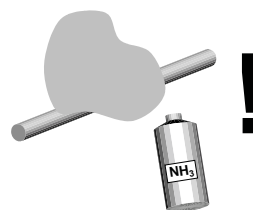
Warning

Ammonia is irritating to eyes, respiratory system and skin!

When checking the tightness with ammonia, wear tight protective goggles, gloves and clothes.

Checking the tightness with ammonia

1. Open the container valve and connection valve until the pressure has reached 3–4 bar, and immediately close it again.
2. If the pressure does not remain constant, there is a leakage.
3. Open the ammonia water bottle.
4. Gently swing the open bottle, until the ammonia vapour comes out. Slowly move the open ammonia bottle along the gas leading parts.



5. If white mist forms, the header line is leaky.
6. Immediately close the container valves.
7. Depressurise the system: Continue operating the system, until the measuring tube on the dosing system shows no more gas flow.
8. Switch off the motive water pump.
9. Eliminate the leakage: For instance tighten the gland in the connection valve, see 9.1 *Leakage at the connection valve*, or tighten the connections of the copper line.
10. Repeat the tightness check.
11. If no white mist forms, the header line and pressure connection are tight.



Fig. 14 Header line without leakage

12. Open all container valves.
13. Switch on the motive water pump and start the system. See the Installation and Operating Instructions of the gas dosing system.

Checking the tightness of the gas dosing system

The tightness of the entire gas dosing system has to be checked. See the Installation and Operating Instructions of the gas dosing system.

Leakages

Due to the heavy corrosiveness of humid chlorine gas, leakages increase quickly. Therefore, even the smallest leakage must be eliminated immediately.

TM04 8654 4212

TM04 8636 4212

5.2.3 Checking the tightness of parts leading ammonia

Even very small leakages become evident with burning sulphur, which reacts with ammonia forming white mist.

1. Open the container valve and connection valve until the pressure has reached 3-4 bar, and immediately close it again.
2. If the pressure does not remain constant, there is a leakage.
3. Slowly move a burning sulphur strip along the gas leading parts.
4. If white mist forms, the header line is leaky.
5. Immediately close the container valves.
6. Depressurise the system: Continue operating the system, until the measuring tube on the dosing system shows no more gas flow.
7. Switch off the motive water pump.
8. Eliminate the leakage: For instance tighten the gland in the connection valve, see 9.1 *Leakage at the connection valve*, or tighten the connections of the copper line.
9. Repeat the tightness check.
10. If no white mist forms, the header line and pressure connection are tight.
11. Open all container valves.
12. Switch on the motive water pump and start the system.
See the Installation and Operating Instructions of the gas dosing system.

Checking the tightness of the gas dosing system

The tightness of the entire gas dosing system has to be checked. See the Installation and Operating Instructions of the gas dosing system.

5.3 Start-up

Warning

Before starting the header line, have its good condition checked by an expert, see 1.3 Qualification of the users.

Check the tightness only if the entire system is ready for start-up.

Danger of chlorine gas leakage!

Start-up of the header line has to be done together with the gas dosing system. See the Installation and Operating Instructions of the gas dosing system.



6. Operation

The header line doesn't have to be operated.

If no faults occur, for uninterrupted gas extraction after start-up exchange empty gas containers by full ones in good time.

6.1 Container exchange

1. Shut off the respective connection valve by tightening the lock nut (1) as far as it will go, in order to avoid gas escape from the header line.
2. Close the cylinder valve or drum valve.
3. Exchange the empty container by a full one.
4. Connect the full container to the connection valve.
5. Open the cylinder valve or drum valve.
6. Open the connection valve by loosening the lock nut (1) until at least 3 convolutions (12 to 13 mm) become visible between the lock nut and the valve body.
7. Check the tightness of the header line, see section 5.2 *Checking the tightness*.

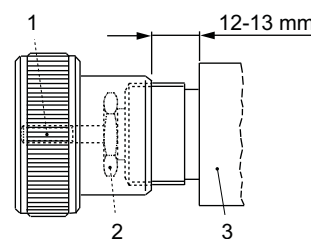


Fig. 15 Connection valve

Pos.	Description
1	Lock nut
2	Gland
3	Valve body

TM04 8609 4112

7. Maintenance

Intervals for cleaning and maintenance

- Check the connection valves at least once a year.
- When exchanging containers, but at least once a year:
 - Clean the sealing surfaces and holes of the connection valves
 - Replace the gaskets of the connection valves
- Replace the flexible copper lines every two years.

Warning

Maintenance has to be carried out only by authorised and qualified personnel.

Before starting cleaning and maintenance work, switch off the entire system.

The system must be pressureless!

Danger of gas escape!

Put on the required protective equipment.



Warning

Make sure that the system cannot be started accidentally while working at it.



Warning

Before re-starting, check the tightness.

Danger of gas escape!



7.1 Replacing the flexible lines

The flexible copper lines have to be replaced every two years.

7.1.1 Flexible copper lines for chlorine

Material	Length [m]	Product No
Copper	1	96688686
	2	96690366

7.1.2 Flexible stainless-steel lines for ammonia and sulphur dioxide

Material	Length [m]	Product No
Stainless steel	1	96688690
	2	96688691

1. Depressurise the system.
 - Close the container valves while the system is still running.
 - Continue operating the system, until the measuring tube on the dosing system shows no more gas flow.
 - Switch off the motive water pump. See the Installation and Operating Instructions of the gas dosing system.
2. Replace the old flexible lines by new ones.

7.2 Retrofitting old header lines

Old header lines have flanged connection screwings. If your system is equipped with old header lines, these have to be retrofitted and the old connection screwings have to be replaced by new soldered connection screwings.

The connection screwings can be tightened with Loctite or hemp.

Note

Mounting the connection screwings with Loctite leads to better stability.

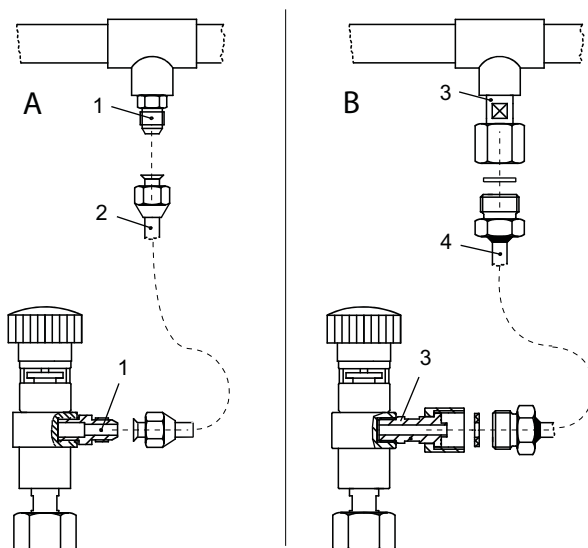


Fig. 16 A: Flanged connection screwing (old)
B: Soldered connection screwing (new)

Pos.	Description
1	Flanged connection screwing
2	Copper line, flanged
3	Soldered connection screwing
4	Copper line, soldered

7.2.1 Assembly with Loctite

Required equipment: Loctite 273, screw tap G 1/4, brush



Warning

Observe safety instructions when handling adhesives!

1. Warm up the joint.
2. Remove the old connection screwing (1).
3. Remove all grease from the joint.
4. Remove adhesive residues from the internal thread with screw tap G 1/4.
5. Apply Loctite to the thread of the connection screwing (3). The thread must be fully wetted.
6. Screw in the connection screwing.
7. Let the joint harden for 24 hours.

7.2.2 Assembly with hemp

Required equipment: Hemp, screw tap G 1/4, brush

1. Warm up the joint.
2. Remove the old connection screwing (1).
3. Remove all grease from the joint.
4. Remove adhesive residues from the internal thread with screw tap G 1/4.
5. Seal the connection screwing properly with hemp and screw in.

TM04 8656 4112

7.3 Maintenance of connection valve

When exchanging containers, but at least once a year, the gaskets of the connection valves must be replaced.

Maintenance kit for connection valves (gaskets)

Set	Valve size	Product No
5 gaskets for connection valves	1 1/4"	95702615
	G 1/2 (copper line)	95717254
	1", G 3/4	96681161
	1.03" - 14, yoke, closed	95707704
	Yoke, open	98026856
	G 5/8, W 21.80x1/14	95707417

Required equipment: Hot water (approx. 40 °C) and a soft brush.

1. Depressurise the system.
2. Dismantle the connection valve.
3. Unscrew the union nuts (16 and 17).
4. Remove the gaskets (10 and 13).
5. Clean and dry the sealing surfaces and holes.

Caution *Danger of corrosion when assembling moist parts!*

6. Insert new gaskets (10 and 13)
7. Slightly grease the gaskets. No grease may get into the gas-leading parts.
8. Connect the flexible line and the gas container.

Assembly drawing

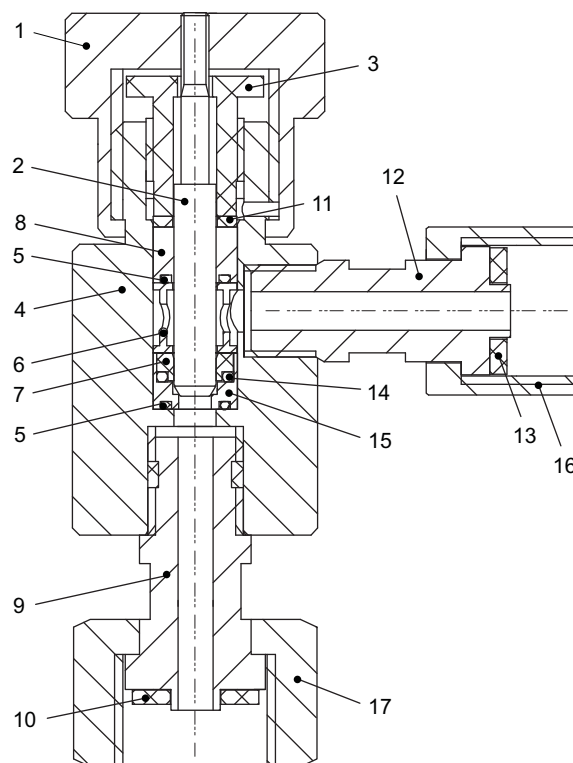


Fig. 17 Connection valve

Pos.	Description
1	Lock nut
2	Spindle
3	Gland
4	Valve body
5	Packing
6	Gland
7	Packing
8	Packing
9	Connection screwing (gas container)
10	Gasket
11	Sliding disk
12	Connection screwing (copper line)
13	Gasket
14	O-ring
15	Support disk
16	Union nut for connection of gas container
17	Union nut for connection of copper line

TM04 8606 4112

7.4 Maintenance of the shut-off valve

Every two years, but at least when leakages occur, the gaskets and packings of the shut-off valve must be replaced.

Set	Shut-off valve	Product No
2 external flat gaskets, 4 internal packings	DN 15	96688716
	DN 25	96688717

Required equipment: Hot water (approx. 40 °C) and a soft brush.

1. Open the shut-off valve completely.
2. Depressurise the system.
3. Unscrew the nuts at the upper part of body (1).
4. Remove the upper part of body with piston (5).
5. Take the packings (3a) and lantern (3) out of the valve body.
6. Clean the valve body and all other parts. Replace damaged or worn parts.

Caution *Danger of corrosion when assembling moist parts!*

7. Dry all parts well.
8. Insert lantern and new packings.
9. Place upper part of body with piston.
10. Screw on the nuts slightly by hand.
11. Close the shut-off valve.
12. Tighten the nuts with an open-end wrench.

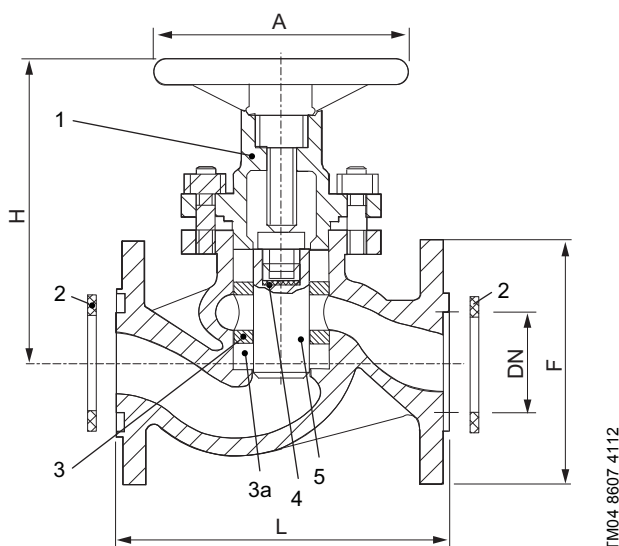


Fig. 18 Shut-off valve

Pos.	Description	DN [mm]	F [mm]	L [mm]	H [mm]	A [mm]	Weight [kg]	Product number
1	Upper part of body	15	95	130	105	100	2.5	96706445
2	Flat gasket	25	115	150	120	120	4.15	96706446
3	Lantern							
3a	Packing							
4	Pressure disc							
5	Piston							

8. Spare parts

Flexible copper lines for Cl₂

Material	Length [m]	Product No
Copper	1	96688686
	2	96690366

Flexible stainless-steel lines for NH₃ und SO₂

Material	Length [m]	Product No
Stainless steel	1	96688690
	2	96688691

Connection valves

Set	Valve size	Product No
5 gaskets for connection valves	1 1/4"	95702615
	G 1/2 (copper line)	95717254
	1", G 3/4	96681161
	1.03" - 14, yoke, closed	95707704
	Yoke, open	98026856
	G 5/8, W 21.80x1/14	95707417

Shut-off valves

Set	Shut-off valve	Product No
2 external flat gaskets, 4 internal packings	DN 15	96688716
	DN 25	96688717

9. Fault finding

Possible faults at the header line:

- Leakages at the connecting parts between copper line and connection valve of the gas container.
- Leakages at the connecting part between copper line and steel line.

9.1 Leakage at the connection valve

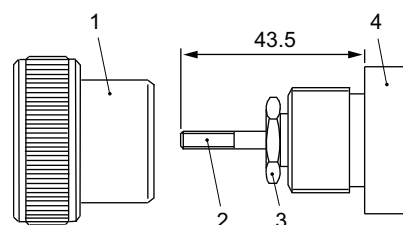


Fig. 19 Connection valve

Pos.	Description
1	Lock nut
2	Spindle
3	Gland
4	Valve body

1. Unscrew lock nut (1).
2. Tighten the gland (3) with a torque of 15 Nm.
3. Retighten lock nut.
4. If the leakage still exists, install a new connection valve.

9.2 Leakage at the shut-off valve

See 7.4 Maintenance of the shut-off valve.

10. Decommissioning

Warning

Decommissioning must be carried out by authorised and qualified personnel.

Before dismantling, the entire system must be switched off.

The system must be pressureless.

Danger of gas escape!

Put on the required protective equipment.



Note

If possible flush the system with nitrogen.

10.1 Short-term decommissioning (up to 2 weeks)

1. Close the connecting valves of all drums or cylinders.
2. Start the system and stop it when the variable area flowmeter indicates no more flow.
3. Close the adjusting valve at the gas dosing unit.
4. Stop the motive water pump or close the solenoid valve.

10.2 Long-term decommissioning (more than 2 weeks)

1. Close the connecting valves of all drums or cylinders.
2. Start the system and stop it when the variable area flowmeter indicates no more flow.
3. Close the adjusting valve at the gas dosing unit.
4. Stop the motive water pump or close the solenoid valve.
5. Close the shut-off valves of the motive water pump.
6. Close the shut-off valves upstream and downstream of the injector.
7. Close the shut-off valve of the injection unit.
8. Remove and dismantle the header line.
9. Use warm water to clean the parts that have been in contact with chlorine.
10. Dry all parts completely.

Caution

Seal up the connections and valves to make sure, that no moisture can ingress!

11. Store all components of the header line.

11. Disposal

This product or parts of it must be disposed of in an environmentally sound way. Use appropriate waste collection services. If this is not possible, contact the nearest Grundfos company or service workshop.

Subject to alterations.

Argentina

Bombas GRUNDFOS de Argentina S.A.
Ruta Panamericana km. 37.500 Centro
Industrial Garin
1619 - Garin Pcia. de B.A.
Phone: +54-3327 414 444
Telefax: +54-3327 411 111

Australia

GRUNDFOS Pumps Pty. Ltd.
P.O. Box 2040
Regency Park
South Australia 5942
Phone: +61-8-8461-4611
Telefax: +61-8-8340 0155

Austria

GRUNDFOS Pumpen Vertrieb Ges.m.b.H.
Grundfosstraße 2
A-5082 Grödig/Salzburg
Tel.: +43-6246-883-0
Telefax: +43-6246-883-30

Belgium

N.V. GRUNDFOS Bellux S.A.
Boomsesteenweg 81-83
B-2630 Aartselaar
Tel.: +32-3-870 7300
Télécopie: +32-3-870 7301

Belarus

Представительство ГРУНДФОС в
Минске
220125, Минск
ул. Шафарнянская, 11, оф. 56
Tel.: +7 (375 17) 286 39 72, 286 39 73
Факс: +7 (375 17) 286 39 71
E-mail: minsk@grundfos.com

Bosnia/Herzegovina

GRUNDFOS Sarajevo
Trg Heroja 16,
BiH-71000 Sarajevo
Phone: +387 33 713 290
Telefax: +387 33 659 079
e-mail: grundfos@bih.net.ba

Brazil

BOMBAS GRUNDFOS DO BRASIL
Av. Humberto de Alencar Castelo Branco,
630
CEP 09850 - 300
São Bernardo do Campo - SP
Phone: +55-11 4393 5533
Telefax: +55-11 4343 5015

Bulgaria

GRUNDFOS Bulgaria EOOD
Slatina District
Iztochna Tangenta street no. 100
BG - 1592 Sofia
Tel. +359 2 49 22 200
Fax. +359 2 49 22 201
email: bulgaria@grundfos.bg

Canada

GRUNDFOS Canada Inc.
2941 Brighton Road
Oakville, Ontario
L6H 6C9
Phone: +1-905 829 9533
Telefax: +1-905 829 9512

China

Grundfos Alldos
Dosing & Disinfection
ALLDOS (Shanghai) Water Technology
Co. Ltd.
West Unit, 1 Floor, No. 2 Building (T 4-2)
278 Jinhu Road, Jin Qiao Export Process-
ing Zone
Pudong New Area
Shanghai, 201206
Phone: +86 21 5055 1012
Telefax: +86 21 5032 0596
E-mail: grundfosalldos-CN@grundfos.com

China

GRUNDFOS Pumps (Shanghai) Co. Ltd.
50/F Maxdo Centre No. 8 Xing Yi Rd.
Hongqiao Development Zone
Shanghai 200336
PRC
Phone: +86-21 6122 5222
Telefax: +86-21 6122 5333

Croatia

GRUNDFOS CROATIA d.o.o.
Cebini 37, Buzin
HR-10010 Zagreb
Phone: +385 1 6595 400
Telefax: +385 1 6595 499
www.hr.grundfos.com

Czech Republic

GRUNDFOS s.r.o.
Čapkovského 21
779 00 Olomouc
Phone: +420-585-716 111
Telefax: +420-585-716 299

Denmark

GRUNDFOS DK A/S
Martin Bachs Vej 3
DK-8850 Bjerringbro
Tlf.: +45-87 50 50 50
Telefax: +45-87 50 51 51
E-mail: info_GDK@grundfos.com
www.grundfos.com/DK

Estonia

GRUNDFOS Pumps Eesti OÜ
Peterburi tee 92G
11415 Tallinn
Tel: + 372 606 1690
Fax: + 372 606 1691

Finland

OY GRUNDFOS Pumput AB
Mestarintie 11
FIN-01730 Vantaa
Phone: +358-(0)207 889 900
Telefax: +358-(0)207 889 550

France

Pompes GRUNDFOS Distribution S.A.
Parc d'Activités de Chesnes
57, rue de Malacombe
F-38290 St. Quentin Fallavier (Lyon)
Tél.: +33-4 74 82 15 15
Télécopie: +33-4 74 94 10 51

Germany

GRUNDFOS Water Treatment GmbH
Reetzstraße 85
D-76327 Pfingsttal (Söllingen)
Tel.: +49 7240 61-0
Telefax: +49 7240 61-177
E-mail: gwt@grundfos.com

Germany

GRUNDFOS GMBH
Schlüterstr. 33
40699 Erkrath
Tel.: +49-(0) 211 929 69-0
Telefax: +49-(0) 211 929 69-3799
E-mail: infoservice@grundfos.de
Service in Deutschland:
E-mail: kundendienst@grundfos.de

Greece

GRUNDFOS Hellas A.E.B.E.
20th km. Athinon-Markopoulou Av.
P.O. Box 71
GR-19002 Peania
Phone: +0030-210-66 83 400
Telefax: +0030-210-66 46 273

Hong Kong

GRUNDFOS Pumps (Hong Kong) Ltd.
Unit 1, Ground floor
Siu Wai Industrial Centre
29-33 Wing Hong Street &
68 King Lam Street, Cheung Sha Wan
Kowloon
Phone: +852-27861706 / 27861741
Telefax: +852-27858664

Hungary

GRUNDFOS Hungária Kft.
Park u. 8
H-2045 Törökbalint,
Phone: +36-23 511 110
Telefax: +36-23 511 111

India

GRUNDFOS Pumps India Private Limited
118 Old Mahabalipuram Road
Thoraiakkam
Chennai 600 097
Phone: +91-44 4596 6800

Indonesia

PT GRUNDFOS Pompa
Jl. Rawa Sumur III, Blok III / CC-1
Kawasan Industri, Pulogadung
Jakarta 13930
Phone: +62-21-460 6909
Telefax: +62-21-460 6910 / 460 6901

Ireland

GRUNDFOS (Ireland) Ltd.
Unit A, Merrywell Business Park
Ballymount Road Lower
Dublin 12
Phone: +353-1-4089 800
Telefax: +353-1-4089 830

Italy

GRUNDFOS Pompe Italia S.r.l.
Via Gran Sasso 4
I-20060 Truccazzano (Milano)
Tel.: +39-02-95838112
Telefax: +39-02-95309290 / 95838461

Japan

GRUNDFOS Pumps K.K.
Gotanda Metalion Bldg. 5F,
5-21-15, Higashi-gotanda
Shiagawa-ku, Tokyo,
141-0022 Japan
Phone: +81 35 448 1391
Telefax: +81 35 448 9619

Korea

GRUNDFOS Pumps Korea Ltd.
6th Floor, Aju Building 679-5
Yeoksam-dong, Kangnam-ku, 135-916
Seoul, Korea
Phone: +82-2-5317 600
Telefax: +82-2-5633 725

Latvia

SIA GRUNDFOS Pumps Latvia
Deglava biznesa centrs
Augusta Deglava ielā 60, LV-1035, Rīga,
Tālr.: + 371 714 9640, 7 149 641
Fakss: + 371 914 9646

Lithuania

GRUNDFOS Pumps UAB
Smolensko g. 6
LT-03201 Vilnius
Tel: + 370 52 395 430
Fax: + 370 52 395 431

Malaysia

GRUNDFOS Pumps Sdn. Bhd.
7 Jalan Peguam U1/25
Glenmarie Industrial Park
40150 Shah Alam
Selangor
Phone: +60-3-5569 2922
Telefax: +60-3-5569 2866

Mexico

Bombas GRUNDFOS de México S.A. de
C.V.
Boulevard TLC No. 15
Parque Industrial Stiva Aeropuerto
Apodaca, N.L. 66600
Phone: +52-81-8144 4000
Telefax: +52-81-8144 4010

Netherlands

GRUNDFOS Netherlands
Veluwezoom 35
1326 AE Almere
Postbus 22015
1302 CA ALMERE
Tel.: +31-88-478 6336
Telefax: +31-88-478 6332
E-mail: info_gnl@grundfos.com

New Zealand

GRUNDFOS Pumps NZ Ltd.
17 Beatrice Tinsley Crescent
North Harbour Industrial Estate
Albany, Auckland
Phone: +64-9-415 3240
Telefax: +64-9-415 3250

Norway

GRUNDFOS Pumper A/S
Strømsveien 344
Postboks 235, Leirdal
N-1011 Oslo
Tlf.: +47-22 90 47 00
Telefax: +47-22 32 21 50

Poland

GRUNDFOS Pompy Sp. z o.o.
ul. Klonowa 23
Baranowo k. Poznania
PL-62-081 Przeźmierowo
Tel.: (+48-61) 650 13 00
Fax: (+48-61) 650 13 50

Portugal

Bombas GRUNDFOS Portugal, S.A.
Rua Calvet de Magalhães, 241
Apartado 1079
P-2770-153 Paço de Arcos
Tel.: +351-21-440 76 00
Telefax: +351-21-440 76 90

Romania

GRUNDFOS Pompe România SRL
Bd. Biruintei, nr 103
Pantelimon county Ilfov
Phone: +40 21 200 4100
Telefax: +40 21 200 4101
E-mail: romania@grundfos.ro

Russia

ООО Грундфос
Россия, 109544 Москва, ул. Школьная
39
Тел. (+7) 495 737 30 00, 564 88 00
Факс (+7) 495 737 75 36, 564 88 11
E-mail grundfos.moscow@grundfos.com

Serbia

GRUNDFOS Predstavništvo Beograd
Dr. Milutina Ivkovića 2a/29
YU-11000 Beograd
Phone: +381 11 26 47 877 / 11 26 47 496
Telefax: +381 11 26 48 340

Singapore

GRUNDFOS (Singapore) Pte. Ltd.
25 Jalan Tukang
Singapore 619264
Phone: +65-6681 9688
Telefax: +65-6681 9689

Slovakia

GRUNDFOS s.r.o.
Prievozská 4D
821 09 BRATISLAVA
Phona: +421 2 5020 1426
sk.grundfos.com

Slovenia

GRUNDFOS d.o.o.
Šlandrova 8b, SI-1231 Ljubljana-Črnuče
Phone: +386 1 568 0610
Telefax: +386 1 568 0619
E-mail: slovenia@grundfos.si

South Africa

Grundfos (PTY) Ltd.
Corner Mountjoy and George Allen Roads
Wilbart Ext. 2
Bedfordview 2008
Phone: (+27) 11 579 4800
Fax: (+27) 11 455 6066
E-mail: lsmart@grundfos.com

Spain

Bombas GRUNDFOS España S.A.
Camino de la Fuentecilla, s/n
E-28110 Algiete (Madrid)
Tel.: +34-91-848 8800
Telefax: +34-91-628 0465

Sweden

GRUNDFOS AB
(Box 333) Lunnagårdsgatan 6
431 24 Mölndal
Tel.: +46 31 332 23 000
Telefax: +46 31-331 94 60

Switzerland

GRUNDFOS ALLDOS International AG
Schönmattdstraße 4
CH-4153 Reinach
Tel.: +41-61-717 5555
Telefax: +41-61-717 5500
E-mail: grundfosalldos-CH@grundfos.com

Switzerland

GRUNDFOS Pumpen AG
Bruggacherstrasse 10
CH-8117 Fällanden/ZH
Tel.: +41-44-806 8111
Telefax: +41-44-806 8115

Taiwan

GRUNDFOS Pumps (Taiwan) Ltd.
7 Floor, 219 Min-Chuan Road
Taichung, Taiwan, R.O.C.
Phone: +886-4-2305 0868
Telefax: +886-4-2305 0878

Thailand

GRUNDFOS (Thailand) Ltd.
92 Chaloom Phrakiat Rama 9 Road,
Dokmai, Pravej, Bangkok 10250
Phone: +66-2-725 8999
Telefax: +66-2-725 8998

Turkey

GRUNDFOS POMPA San. ve Tic. Ltd. Sti.
Gebze Organize Sanayi Bölgesi
İhsan dede Caddesi,
2. yol 200. Sokak No. 204
41490 Gebze/ Kocaeli
Phone: +90 - 262-679 7979
Telefax: +90 - 262-679 7905
E-mail: satis@grundfos.com

Ukraine

Бізнес Центр Європа
Столичне шосе, 103
м. Київ, 03131, Україна
Телефон: (+38 044) 237 04 00
Факс: (+38 044) 237 04 01
E-mail: ukraine@grundfos.com

United Arab Emirates

GRUNDFOS Gulf Distribution
P.O. Box 16768
Jebel Ali Free Zone
Dubai
Phone: +971-4- 8815 166
Telefax: +971-4-8815 136

United Kingdom

GRUNDFOS Pumps Ltd.
Grovebury Road
Leighton Buzzard/Beds. LU7 4TL
Phone: +44-1525-850000
Telefax: +44-1525-850011

U.S.A.

GRUNDFOS Pumps Corporation
17100 West 118th Terrace
Olathe, Kansas 66061
Phone: +1-913-227-3400
Telefax: +1-913-227-3500

Uzbekistan

Grundfos Tashkent, Uzbekistan The Repre-
sentative Office of Grundfos Kazakhstan in
Uzbekistan
38a, Oybek street, Tashkent
Телефон: (+998) 71 150 3290 / 71 150
3291
Факс: (+998) 71 150 3292

Addresses revised 21.05.2014

98401778 1114
ECM: 1105371

The name Grundfos, the Grundfos logo, and be think innovate are registered trademarks owned by Grundfos Holding A/S or Grundfos A/S, Denmark. All rights reserved worldwide. © Copyright Grundfos Holding A/S